INTRO TO CYBERSECURITY AND HACKING

EESTEC Academy 2026; Workshop Connected & Secure

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Information & Systems Security Group

ISSG

Information & Systems Security Group is dedicated to advancing cybersecurity through research and development.

https://issg.gr

Research Topics

- Identity Management
- Network Security
- Hardware Security
- Artificial Intelligence
- And much more

Activities

- HackingGroup
- HTB Meetups
- Competitions

Agenda

Foundations

- Intro Cybersecurity
- Intro Hacking
- Career paths

Practice

- How hacking works
- Linux basics
- Common tools

Hands-on

- CTF workflow
- Demo Hacking
- Resources for practice

Presentation Location

Presentation

Github: nikolasfil/Cybersecurity-Wiki/Events/EESTEC/Academy-2026/presentation.pdf

Introduction to Cybersecurity

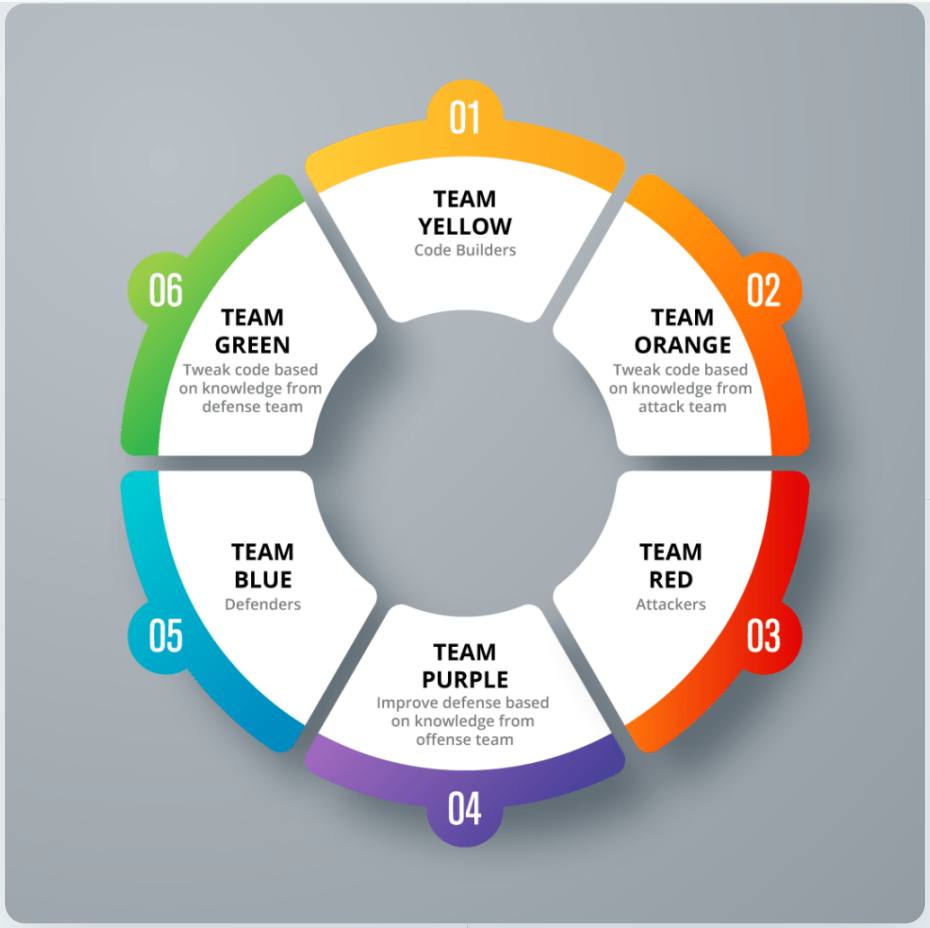
Cybersecurity is the practice of protecting systems, networks, applications, and data from unauthorized access, misuse, disruption, or destruction.

It is not only about "stopping hackers". It is about understanding how technology fails, how people make mistakes, and how organizations reduce risk.

Goal	Meaning	Example
Confidentiality	Only the right people can access information	Passwords, encryption, access control
Integrity	Data and systems stay correct and trustworthy	Checksums, logs, code signing
Availability	Services remain usable when needed	Backups, redundancy, DDoS protection

From an attacker's perspective, we ask: **How can this system be abused?**
From a defender's perspective, we ask: **How do we prevent, detect, and respond?**

Cybersecurity and Colors



Cybersecurity Roles

Source: UNIHACK 2026 / ECSC national team presentation image.

Career Paths

Blue Team

- SOC analyst
- Incident responder
- Detection engineer

Red Team

- Pentester
- Red team operator
- Exploit developer

Builder Roles

- AppSec engineer
- Cloud security
- Security researcher

Core Skills Across All Paths

Linux, networking, scripting, web basics, curiosity, and strong documentation habits.

Introduction to Hacking

HACKING CULTURE

> "Hacking is the art of exploiting weaknesses in a system to achieve a goal."

— Jon Erickson, *Hacking: The Art of Exploitation*

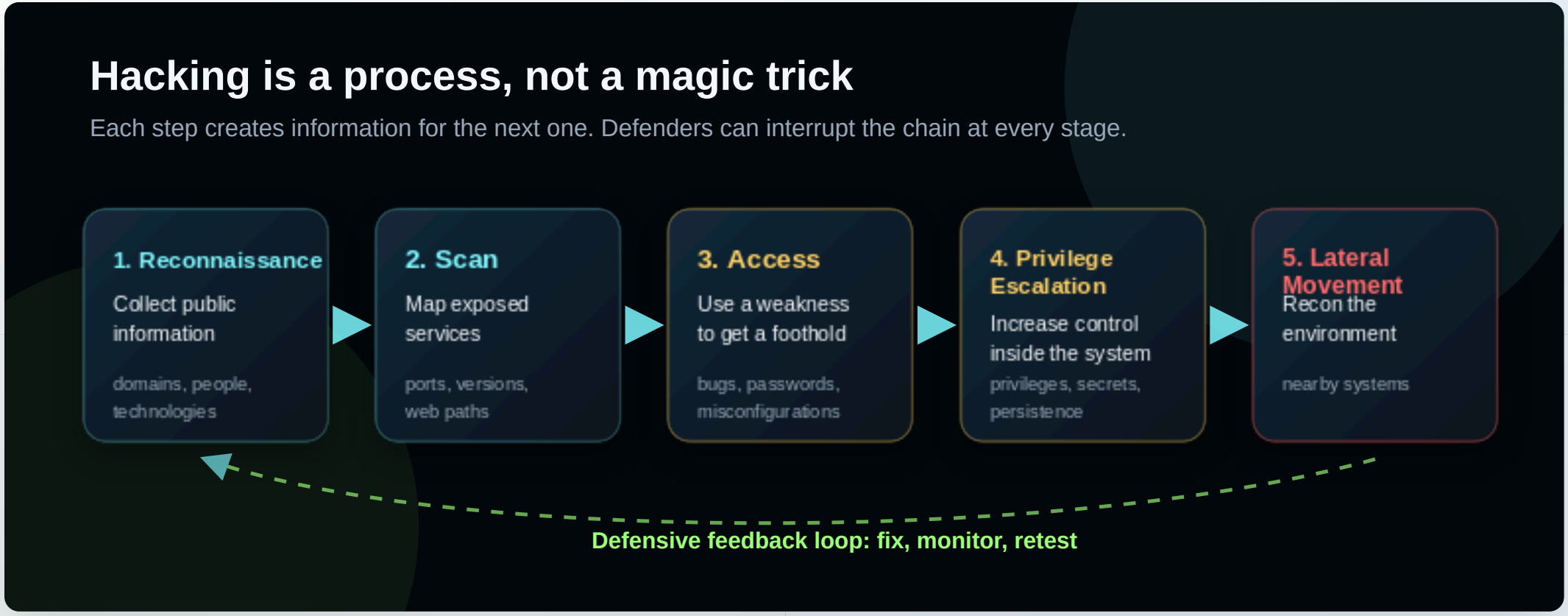
> "A hacker is someone who enjoys exploring the details of programmable systems and stretching their capabilities"

— Charlie Miller, *The Hacker's Handbook Series*

Hacking Hats

The real difference is permission, intent, and impact.

Hacking Steps



Useful Linux Commands

Generic Commands

Command	Meaning
`ls`	list files
`cd`	change directory
`pwd`	show current location
`cat`	print file
`less`	view large files
`cp`	copy files
`mv`	move / rename
`rm`	delete
`man`	help

Useful in Hacking

Command	Use
`grep`	find secrets or interesting text
`find`	locate sensitive files
`ps`	inspect processes
`ss`	discover open ports
`curl`	interact with web services
`chmod`	change permissions
`history`	find previously typed commands
`python3`	scripting and payloads
`whoami`	identify the current user

Hacking Tools

Discovery

- `nmap`
- `gobuster`
- `ffuf`

Web & Data

- Burp Suite
- sqlmap
- CyberChef

Systems

- Wireshark
- Metasploit
- LinPEAS

Tools matter less than knowing what question each tool is answering.

nmap

Network Mapper is used to discover hosts, open ports, running services, and operating systems on a network.

Common uses

- Discover live hosts
- Identify open ports
- Detect services and versions
- OS fingerprinting

```
 - Check if an IP address is up, and guess the remote hosts operating system:
   nmap -O ip_or_hostname

 - Try to determine whether the specified hosts are up (ping scan) and what their
names and MAC addresses are:
   sudo nmap -sn ip_or_hostname optional_another_address

 - Also enable scripts, service detection, OS fingerprinting and traceroute:
   nmap -A address_or_addresses

 - Scan a specific list of ports (use '-p-' for all ports from 1 to 65535):
   nmap -p port1,port2,...,portN address_or_addresses

 - Perform service and version detection of the top 1000 ports using default NSE
scripts; writing results ('-oN') to output file:
   nmap -sC -sV -oN top-1000-ports.txt address_or_addresses

 - Scan target(s) carefully using 'default and safe' NSE scripts:
   nmap --script "default and safe" address_or_addresses

 - Scan web server running on standard ports 80 and 443 using all available 'http-
*' NSE scripts:
   nmap --script "http-*" address_or_addresses -p 80,443

 - Perform a stealthy very slow scan ('-T0') trying to avoid detection by IDS/IPS
and use decoy ('-D') source IP addresses:
   nmap -T0 -D decoy1_ipaddress,decoy2_ipaddress,...,decoyN_ipaddress
```

gobuster

Gobuster is used for brute-forcing directories, DNS subdomains, virtual hosts, and other targets.

Common uses

- Directory enumeration
- DNS subdomain enumeration
- Fuzzing
- S3 bucket enumeration
- VHost enumeration

```
- Discover directories and files that match in the wordlist:
  gobuster dir --url https://example.com/ --wordlist path/to/file

- Discover subdomains:
  gobuster dns --domain example.com --wordlist path/to/file

- Discover Amazon S3 buckets:
  gobuster s3 --wordlist path/to/file

- Discover other virtual hosts on the server:
  gobuster vhost --url https://example.com/ --wordlist path/to/file

- Fuzz the value of a parameter:
  gobuster fuzz --url https://example.com/?parameter=FUZZ --wordlist
path/to/file

- Fuzz the name of a parameter:
  gobuster fuzz --url https://example.com/?FUZZ=value --wordlist path/to/file
```

ffuf

ffuf is a fast web fuzzer written in Go for directory discovery, virtual host discovery, and parameter fuzzing.

Common uses

- Directory enumeration
- DNS subdomain enumeration
- GET and POST parameter fuzzing
- VHost discovery

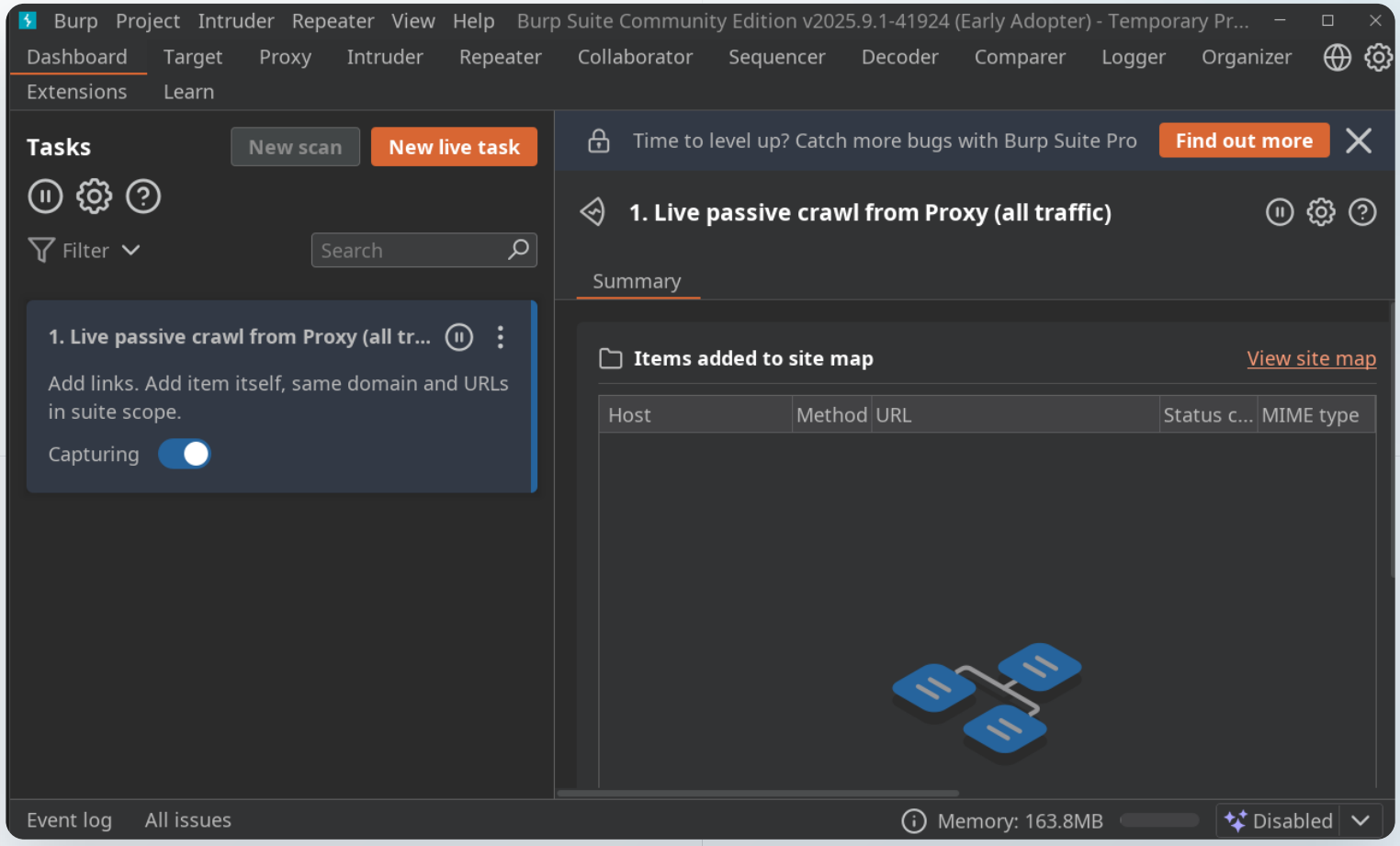
```
  - Enumerate directories using [c]olored output and a [w]ordlist specifying a
target [u]RL:
    ffuf -c -w path/to/wordlist.txt -u https://example.com/FUZZ

  - Enumerate webservers of subdomains by changing the position of the keyword:
    ffuf -w path/to/subdomains.txt -u https://FUZZ.example.com

  - Fuzz with specified [t]hreads (default: 40) and pro[x]ying the traffic and
save [o]utput to a file:
    ffuf -o -w path/to/wordlist.txt -u https://example.com/FUZZ -t 500 -x http://12
7.0.0.1:8080

  - Fuzz a specific [H]eader ("Name: Value") and [m]atch HTTP status [c]odes:
    ffuf -w path/to/wordlist.txt -u https://example.com -H "Host: FUZZ" -mc 200

  - Fuzz with specified HTTP method and [d]ata, while [f]iltering out specific
status [c]odes and response [s]ize:
    ffuf -w path/to/postdata.txt -X POST -d "username=admin\&password=FUZZ" -u
https://example.com/login.php -fc 302,401-499 -fs 1234

  - Fuzz multiple positions with multiple wordlists using different modes and
[a]uto [c]alibration to reduce false positives:
    ffuf -w path/to/keys:KEY -w path/to/values:VALUE -mode pitchfork|clusterbomb -
u https://example.com/id?KEY=VALUE -ac

  - Proxy requests through a HTTP MITM pro[x]y (such as Burp Suite or mitmproxy):
    ffuf -w path/to/wordlist -x http://127.0.0.1:8080 -u https://example.com/FUZZ
```

Burp Suite

Burp Suite is a web security testing platform used for intercepting, modifying, and analyzing HTTP traffic.

Common uses

- Intercept web traffic
- Modify requests
- Test authentication flows
- Find vulnerabilities



sqlmap

sqlmap is an automated tool used to detect and exploit SQL injection vulnerabilities.

Common uses

- Detect SQL injection
- Dump databases
- Extract credentials
- Automate exploitation

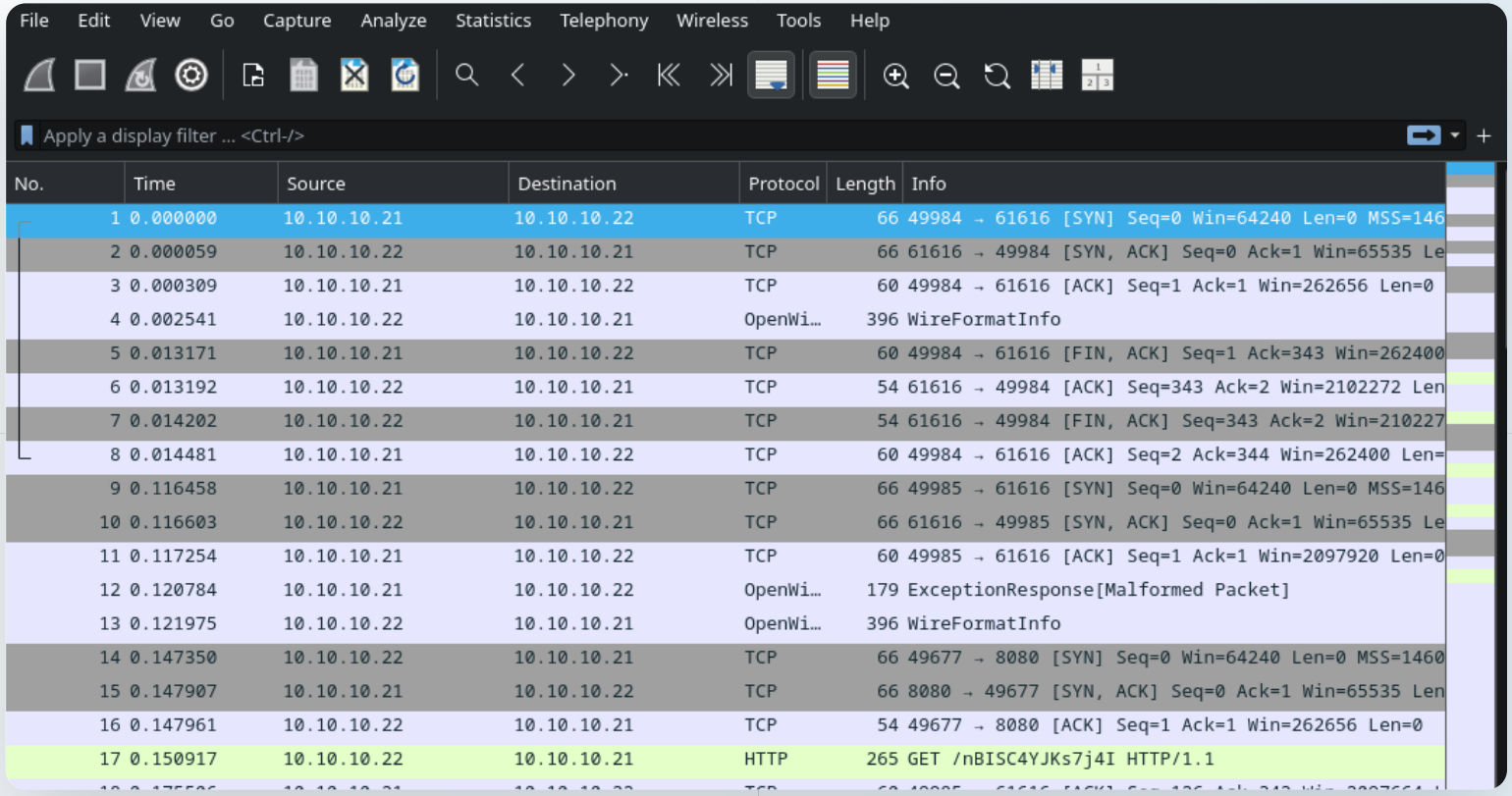
```
  Detect and exploit SQL injection flaws.
  More information: https://github.com/sqlmapproject/sqlmap/wiki/Usage.

  - Run sqlmap against a single target URL:
    python sqlmap.py --url "http://www.example.com/vuln.php?id=1"

  - Send data in a POST request (--data implies POST request):
    python sqlmap.py --url "http://www.example.com/vuln.php" --data="id=1"

  - Change the parameter delimiter (& is the default):
    python sqlmap.py --url "http://www.example.com/vuln.php" --
data="query=foobar;id=1" --param-del=";"

  - Select a random User-Agent from ./txt/user-agents.txt and use it:
    python sqlmap.py --url "http://www.example.com/vuln.php" --random-agent

  - Provide user credentials for HTTP protocol authentication:
    python sqlmap.py --url "http://www.example.com/vuln.php" --auth-type Basic --
auth-cred "testuser:testpass"
```

Wireshark

Wireshark is a network protocol analyzer used to inspect and analyze network traffic.

Common uses

- Packet inspection
- Credential discovery
- Protocol analysis
- Traffic investigation



CyberChef

CyberChef is a browser-based tool for transforming, decoding, encoding, and analyzing data. It is useful in CTFs, incident response, malware analysis, web testing, and any situation where data appears in an unfamiliar format.

Common uses

- Decode Base64, URL encoding, hex, binary, and other formats
- Chain many transformations into one repeatable recipe
- Analyze hashes, timestamps, JWTs, certificates, and compressed data
- Clean up messy input and quickly test hypotheses

Example workflow

Take a suspicious string, paste it into CyberChef, try common decoders, and build a recipe until the data becomes readable.

Metasploit

Metasploit is a penetration testing framework used to find, configure, and run exploit modules against vulnerable systems.

Common uses

- Search for known exploits by service, product, or CVE
- Configure exploit options such as target host, port, and payload
- Launch controlled exploitation in a legal lab environment
- Manage sessions after gaining access to a test machine

Example workflow

Scan a target with `nmap`, identify a vulnerable service, search for a matching Metasploit module, configure `RHOSTS`, choose a payload, and run the module.

Metasploit Commands

Core Commands

Command	Meaning
`msfconsole`	start Metasploit
`search`	search for modules
`use`	select a module
`info`	show module details
`show options`	list required settings
`set RHOSTS`	set target host
`run` / `exploit`	launch the module
`back`	return to previous menu

Post-Exploitation

Command	Use
`sessions`	list active sessions
`background`	background current session
`sysinfo`	show system information
`getuid`	show current user
`shell`	drop to system shell
`upload` / `download`	transfer files

LinPEAS

LinPEAS is a Linux privilege escalation auditing script. After getting access to a Linux machine in a lab or CTF, it checks the system for misconfigurations, weak permissions, interesting files, credentials, vulnerable services, and common privilege escalation paths.

Common uses

- Enumerate users, groups, processes, services, and network connections
- Find readable secrets, SSH keys, passwords, and configuration files
- Detect weak file permissions and writable paths
- Suggest possible Linux privilege escalation techniques

Example workflow

Upload `linpeas.sh` to the target machine, run it as the current user, review the highlighted findings, and manually verify anything that looks exploitable.

Other Useful Tools

- **Ghidra:** reverse engineering framework for compiled binaries
- **revshells.com:** reverse shell command generator
- **pwncat:** advanced reverse shell handling
- **pwntools:** Python library for exploitation
- **gdb:** debugger for reversing, pwn, and binary challenges

- **Wordlists:** https://github.com/kkrypt0nn/wordlists
- **Pentestmonkey:** https://pentestmonkey.net

Intro to CTF

CTF means **Capture The Flag** — a safe, legal way to practice cybersecurity skills through isolated challenges. Each challenge hides a **flag** string (e.g. `flag{s0m3T3xt}`) that you submit to score points. The goal is to find the flag by understanding a vulnerability, decoding data, or reversing a program.

Web	**Cryptography**	**Reverse Engineering**
Forensics	**Binary**	**Miscellaneous**

Short Break

CTF Platform

CyLab Security Academy

- Open the platform
- Register Account

CTF Demonstration

- **Scavenger Hunt**
- **Trickster**
- **SQLite**

Machine Demonstration

Vulnerable Machine

Get the VPN Profile
Follow the instruction for the machine's IP

The Machine deployed is the vulnerable Metasploitable 2

Πιθανόν να έχει διορθωθεί σε εγκατάσταση του 2024, αλλά αξίζει να σημειωθεί.

Η Ομάδα adm — Τι Μπορούμε να Διαβάσουμε;

Το να ανήκουμε στην ομάδα adm μας δίνει πρόσβαση στα logs συστήματος. Το LinPEAS βρήκε τα logs του nginx στο /var/log/nginx/. Σε πραγματική αποστολή, τα logs μπορεί να περιέχουν

Demo Map

Part	Target	Tools
Network exploitation	**Metasploitable 2**	`nmap`, Metasploit, LinPEAS
Web enumeration	**CyLab: Scavenger Hunt**	Gobuster
Request manipulation	**CyLab: GET aHEAD / Who are you? / Cookies**	Burp Suite
File upload + discovery	**CyLab: Trickster**	Burp Suite + Gobuster
SQL injection automation	**CyLab: Irish Name Repo / SQLiLite**	Burp Suite + SQLMap

Resources

General resources: nikolasfil/Cybersecurity-Wiki/wiki/resources

- HackTheBox
- TryHackMe
- CyLab Security Academy
- OverTheWire

- BugBountyHunter
- CyberDefenders
- PortSwigger Web Security Academy
- Hacktricks

QUESTIONS?

Want to learn more ?

HTB Meetup Patras

by ISSG

- Enterprise Platform
- Workshops
- Discussions
- Networking
- Monthly Meetings
- Today 18:00 - 21:00, Hλ4
- link

THANK YOU FOR YOUR ATTENTION

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